

Daily Logic Drill #5 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Answers and Methods.

Section A Rapid Recognition

1. “ $P \Rightarrow Q$, and Q is true; conclude P is true.” Is this reasoning valid?

Answer: No — this is the fallacy of affirming the consequent.

Method: $P \Rightarrow Q$ says nothing about what happens when Q is true; Q could be true for reasons unrelated to P . Concretely: “ n prime $\Rightarrow n$ odd,” and 9 is odd, but 9 is not prime.

Investigate further: This is the single most common logical error on this sheet — watch for it disguised inside longer arguments in Sections B and C.

2. “ $P \Rightarrow Q$, and Q is false; conclude P is false.” Is this reasoning valid?

Answer: Yes — this is modus tollens.

Method: This is exactly proof by contrapositive (Day 3): “not Q , therefore not P ” is the contrapositive of $P \Rightarrow Q$, which is logically equivalent to the original (Day 3, A9).

Investigate further: Modus tollens and “affirming the consequent” (A1) look superficially similar — both start from information about Q — but only one direction is valid.

3. “ $P \Rightarrow Q$, and P is false; conclude Q is false.” Is this reasoning valid?

Answer: No — this is the fallacy of denying the antecedent.

Method: $P \Rightarrow Q$ says nothing about what happens when P is false; Q can still hold for other reasons. Concretely: “ n divisible by 4 $\Rightarrow n$ even,” and 6 is not divisible by 4, but 6 is still even.

Investigate further: This is the mirror image of A1: A1 wrongly runs the implication backwards, this wrongly runs its inverse forwards.

4. “ $P \Rightarrow Q$, and P is true; conclude Q is true.” Is this reasoning valid?

Answer: Yes — this is modus ponens, the basic valid rule.

Method: This is the direct definition of what “ $P \Rightarrow Q$ ” means: whenever P holds, Q is guaranteed to hold too.

Investigate further: Modus ponens (A4) and modus tollens (A2) are the two valid rules; affirming the consequent (A1) and denying the antecedent (A3) are the two corresponding fallacies — all four questions here form a matched set.

5. Spot the flaw: “ n is odd $\Rightarrow n^2$ is odd. $n^2 = 49$ is odd. Therefore $n = 7$.”

Answer: Affirming the consequent — n could also be -7 .

Method: From “ n odd $\Rightarrow n^2$ odd” and “ n^2 odd,” the argument wrongly concludes “ n odd,” then wrongly narrows further to the single value 7. Even granting n odd, $(-7)^2 = 49$ too.

Investigate further: Even when a conclusion happens to be a correct possibility, the reasoning that produced it can still be invalid — validity and truth of the conclusion are separate questions.

6. Spot the flaw: “If a shape is a square, it’s a rectangle. This shape is not a square. Therefore it’s not a rectangle.”

Answer: Denying the antecedent — a 2×3 rectangle is a counterexample.

Method: From “square $\Rightarrow$ rectangle” and “not a square,” the argument wrongly concludes “not a rectangle.” A 2×3 rectangle (Day 1, A4) is not a square but is still a rectangle.

Investigate further: This is the exact geometric instance of the general pattern named in A3.

7. True or false: “If $P \Rightarrow Q$ and $Q \Rightarrow P$ are both true, then assuming Q to conclude P becomes valid reasoning.”

Answer: True.

Method: If both directions hold (Day 2’s “iff”), then $Q \Rightarrow P$ is itself one of the given true statements, so concluding P from Q is just modus ponens (A4) applied to $Q \Rightarrow P$ — no longer the fallacy of A1.

Investigate further: “Affirming the consequent” is only ever a fallacy when the converse hasn’t been separately established — Day 2’s necessary-and-sufficient language is exactly what licenses reasoning in both directions safely.

8. Divide both sides of $x^2 = 5x$ by x to get $x = 5$. Which solution got lost?

Answer: $x = 0$.

Method: Dividing by x silently assumes $x \neq 0$. Rearranging instead as $x^2 - 5x = 0 \Rightarrow x(x - 5) = 0$ gives both $x = 0$ and $x = 5$.

Investigate further: The fix is never to divide by an expression that could be zero — factor and use the zero-product property instead, as Section C’s C6 revisits.

9. True or false: “Squaring both sides of an equation can introduce extraneous (fake) solutions.”

Answer: True.

Method: Squaring is not injective on the reals: $x = -2$ and $x = 2$ both satisfy $x^2 = 4$, so squaring an equation like $x = -2$ to get $x^2 = 4$ and then solving $x^2 = 4$ produces the extra, invalid solution $x = 2$.

Investigate further: Any solution obtained after squaring must be checked back in the *original* (unsquared) equation before being accepted.

10. True or false: “Dividing both sides of an equation by a variable expression can lose valid solutions.”

Answer: True.

Method: A8 is a direct instance: dividing $x^2 = 5x$ by x silently discards the valid solution $x = 0$, precisely because that expression could equal zero.

Investigate further: Together, A9 and A10 name the two most common “the algebra looks fine but a solution disappeared or appeared” traps — squaring adds, dividing removes.

Section B Manipulation Drills

1. Spot the flaw: “ n^2 is a multiple of 4. Therefore n is a multiple of 4.”

Answer: Affirming the consequent — $n = 6$ is a counterexample.

Method: The true fact is “ n multiple of 4 $\Rightarrow n^2$ multiple of 4”; the argument affirms the consequent (n^2 multiple of 4) to wrongly conclude the hypothesis. $n = 6$ gives $n^2 = 36$, a multiple of 4, but 6 is not.

Investigate further: This is Day 1’s C8 fact, now correctly named as an instance of affirming the consequent.

2. Spot the flaw: “ $\sqrt{x^2} = x$ for all real x .”

Answer: False for negative x ; the correct identity is $\sqrt{x^2} = |x|$.

Method: For $x = -3$: $\sqrt{x^2} = \sqrt{9} = 3$, but $x = -3$. Since $\sqrt{}$ denotes the non-negative square root by convention, $\sqrt{x^2}$ must always be non-negative, so it can only equal x itself when $x \geq 0$.

Investigate further: This silent assumption of non-negativity resurfaces constantly — B9 and B10 both trade on the same confusion between $\sqrt{}$ (single non-negative value) and “square root of” (two possible values, $\pm$).

3. A “proof” that $a = 0$ for any $a = b$: Line I: $a = b$. Line II: $a^2 = ab$. Line III: $a^2 - b^2 = ab - b^2$. Line IV: $(a - b)(a + b) = b(a - b)$. Line V: dividing both sides by $(a - b)$: $a + b = b$, so $a = 0$. Which line introduces the flaw, and why?

Answer: Line V — dividing by $(a - b)$, which equals 0 since $a = b$.

Method: Lines I–IV are all valid algebraic manipulations. Line V divides both sides of $(a - b)(a + b) = b(a - b)$ by $(a - b)$, but since $a = b$ was given, $a - b = 0$ — an unjustified division by zero, which is exactly the classical “proof” that $1 = 2$.

Investigate further: This is the multi-line version of A8’s single-step warning: division is only ever safe once you’ve confirmed the divisor is nonzero, and here the very hypothesis of the problem guarantees it’s zero.

4. Spot the flaw: “If it is sunny, I wear sunglasses. I am wearing sunglasses. Therefore it is sunny.”

Answer: Affirming the consequent — sunglasses could be worn for other reasons (e.g. driving, indoor style).

Method: From “sunny $\Rightarrow$ sunglasses” and “sunglasses,” the argument wrongly concludes “sunny,” the same invalid pattern as A1 and B1, now in an everyday, non-mathematical setting.

Investigate further: Fallacies aren’t specific to numbers — the same structural error appears everywhere “evidence consistent with a cause” gets mistaken for “proof of that cause.”

5. Spot the flaw: “ $x^2 > 4$. Therefore $x > 2$.”

Answer: Incomplete case-split — the correct conclusion is $x > 2$ or $x < -2$.

Method: $x^2 > 4 \iff |x| > 2 \iff x > 2 \text{ or } x < -2$. The argument silently drops the $x < -2$ branch; $x = -3$ satisfies $x^2 = 9 > 4$ but not $x > 2$.

Investigate further: Any time squaring or absolute value is involved, check whether the “obvious” one-sided conclusion has quietly dropped a symmetric negative case — this exact drop resurfaces in Section C’s C2.

6. Spot the flaw: “If n is a multiple of 6, then n is a multiple of 2 and of 3. n is a multiple of 2. Therefore n is a multiple of 6.”

Answer: Affirming (part of) the consequent — $n = 4$ is a counterexample.

Method: The argument only confirms one of the two conjuncts in the consequent (“multiple of 2”), then wrongly concludes the full hypothesis anyway. $n = 4$ is a multiple of 2 but not of 6.

Investigate further: Partially satisfying a compound consequent is just as invalid as fully satisfying it (B1) — the fallacy doesn’t get weaker by only affirming part of Q .

7. Is there a flaw in this argument? “ n is odd (given). Line 1: $n = 2k + 1$ for some integer k . Line 2: $n^2 = 4k^2 + 4k + 1$. Line 3: $n^2 = 2(2k^2 + 2k) + 1$. Line 4: therefore n^2 is odd.”

Answer: No flaw — this is a correct, complete proof (Day 1, B5).

Method: Each line follows directly from the previous by valid algebra, and the conclusion matches exactly what was claimed. There is no fallacious step here.

Investigate further: Not every presented argument is flawed — correctly identifying a valid proof as valid is just as important a skill as catching an invalid one, and exam questions test both.

8. Spot the flaw: “All multiples of 4 are even. n is even. Therefore n is a multiple of 4.”

Answer: Affirming the consequent — $n = 6$ is a counterexample.

Method: From “multiple of 4 $\Rightarrow$ even” and “even,” the argument wrongly concludes “multiple of 4.” $n = 6$ is even but not a multiple of 4.

Investigate further: The same instance ($n = 6$) that disproved B1 disproves this one too — once you’ve found one good counterexample to a converse-error pattern, check whether it kills several related false claims at once.

9. Spot the flaw: “ $\sqrt{4} = \pm 2$, so $\sqrt{4} = 2$ and $\sqrt{4} = -2$ simultaneously, meaning $2 = -2$.”

Answer: Notation error — $\sqrt{4}$ denotes only the non-negative root, 2; it is never -2 .

Method: By convention, $\sqrt{}$ is the *principal* (non-negative) square root function, so $\sqrt{4} = 2$ exactly, a single value. The “ $\pm$ ” only appears when *solving* an equation like $x^2 = 4$, which has two solutions $x = \pm 2$ — it is never part of the value of $\sqrt{4}$ itself.

Investigate further: Confusing “the square root of 4” (a function, one output) with “a number whose square is 4” (a solution set, two outputs) is the exact same confusion underlying B2’s error.

10. Spot the flaw: “Since $(-2)^2 = 2^2 = 4$, we can cancel the squares to conclude $-2 = 2$.”

Answer: Illegitimate cancellation — squaring is not injective, so equal squares don’t imply equal originals.

Method: $(-2)^2 = 4$ and $2^2 = 4$ are both true, but “cancelling the square” (taking square roots of both sides) only legitimately gives $|-2| = |2|$, i.e. $2 = 2$ — true, but not $-2 = 2$.

Investigate further: This is B5 and C5’s “lost sign” pattern one more time, now dressed up as a fake cancellation rule — squares can always be safely compared for equality, but never safely “rooted” back without absolute values.

Section C Substitution & Structure

1. Consider this argument. Line I: let n be even, so $n = 2k$. Line II: $n^3 = 8k^3$. Line III: $n^3 = 2(4k^3)$. Line IV: therefore n^3 is even and n^3 is a multiple of 8. Which best describes this attempt?

Answer: A) Completely correct.

Method: Each line is valid: $n = 2k \Rightarrow n^3 = (2k)^3 = 8k^3$, and $8k^3 = 2(4k^3)$ shows evenness while $8k^3$ is manifestly 8 times an integer, so both parts of the conclusion follow correctly with no gap.

Investigate further: Not every “spot the flaw” question has a flaw — this is Section C’s version of B7’s lesson, at higher difficulty.

2. “Claim: if n is a positive integer and $n^2 > 100$, then $n > 10$. ‘Proof’: $n^2 > 100 \Rightarrow n > 10$ (taking square roots of both sides).” Is this proof valid? If not, give a counterexample.

Answer: The claim itself is true for positive integers, but the given proof step is invalid in general (it silently uses positivity without saying so); as written for all integers it would fail at $n = -11$.

Method: Taking square roots of $n^2 > 100$ directly gives $|n| > 10$, i.e. $n > 10$ or $n < -10$ — not $n > 10$ alone. The claim is true only because the problem restricts to positive integers, which the “proof” never explicitly uses to rule out the $n < -10$ branch. As stated for all integers, $n = -11$ gives $n^2 = 121 > 100$ but $n > 10$ is false.

Investigate further: This is B5’s exact error, now embedded inside a claim that happens to be true for the intended domain — always check whether a domain restriction is doing silent, unacknowledged work in a proof.

3. “Claim: if a, b are integers with $a \mid b$ and $b \mid a$, then $a = b$.” Spot the flaw and give a counterexample.

Answer: Missing case $a = -b$; e.g. $a = 3, b = -3$.

Method: $3 \mid -3$ and $-3 \mid 3$ both hold, but $3 \neq -3$. The correct conclusion from $a \mid b$ and $b \mid a$ is $a = \pm b$, since each divisibility forces $|a| = |b|$ but not a matching sign.

Investigate further: The same “lost sign” theme as B5, C2, B9, B10 — divisibility, like squaring, only pins down magnitude, never sign, unless the domain is restricted to positive integers.

4. “Claim: for positive real x , if $x + \frac{1}{x} = 2$, then $x = 1$. ‘Proof’: $x + \frac{1}{x} = 2 \Rightarrow x^2 + 1 = 2x \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0 \Rightarrow x = 1$.” Is this proof valid?

Answer: Yes, fully valid.

Method: Multiplying by x is legitimate here because x is given to be a positive real, hence nonzero — no division-by-zero risk. Each subsequent step is a standard, reversible algebraic manipulation, and $(x - 1)^2 = 0$ forces the unique solution $x = 1$.

Investigate further: Contrast with B3 and A8: multiplying or dividing by an expression is only dangerous when that expression *could* be zero — here the problem’s own hypothesis (“positive real x ”) rules that out, so the step is safe.

5. “Claim: if n is an integer and n^2 is divisible by 4, then n is divisible by 4. ‘Proof’: n^2 divisible by 4 $\Rightarrow n^2 = 4m$ for some integer $m \Rightarrow n = 2\sqrt{m}$, and since n is an integer this is always even, hence n is divisible by 4.” Identify exactly where this breaks down, and give a counterexample.

Answer: The final leap from “ n is even” to “ n is divisible by 4” is an unjustified non-sequitur; $n = 6$ is a counterexample.

Method: Even granting that $n = 2\sqrt{m}$ shows n is even, evenness alone does not imply divisibility by 4 (Day 1, C8). $n = 6$: $n^2 = 36 = 4 \times 9$ is divisible by 4, but 6 is not divisible by 4.

Investigate further: This is the same false claim as B1 and B8, but now dressed up inside a longer, more convincing-looking algebraic derivation — length and complexity of an argument are no guarantee of correctness.

6. “Claim: the equation $x^3 = x$ has exactly one real solution. ‘Proof’: $x^3 = x \Rightarrow x^3 - x = 0 \Rightarrow x(x^2 - 1) = 0 \Rightarrow x = 0$ (dividing by $(x^2 - 1)$).” Spot the flaw.

Answer: Dividing by $(x^2 - 1)$ instead of case-splitting on the zero-product property loses the solutions $x = \pm 1$; there are three real solutions, not one.

Method: From $x(x^2 - 1) = 0$, the zero-product property gives $x = 0$ or $x^2 - 1 = 0$ (i.e. $x = 1$ or $x = -1$) — three solutions total. Dividing by $(x^2 - 1)$ silently assumes it is nonzero, exactly the same error as A8, discarding the two solutions where it *is* zero.

Investigate further: Whenever a product equals zero, always case-split via the zero-product property rather than dividing out one of the factors — A8’s warning generalises directly to any factored equation.

7. “Claim: if p is prime and $p > 2$, then $p + 1$ is not prime. ‘Proof’: p is odd (Day 1, B9), so $p + 1$ is even. Since $p + 1 > 2$ is even, it cannot be prime, as the only even prime is 2.” Is this proof valid?

Answer: Yes, fully valid.

Method: $p > 2$ prime $\Rightarrow p$ odd (Day 1, B9) $\Rightarrow p + 1$ even. Since $p > 2$, $p + 1 > 3 > 2$, so $p + 1$ is an even number greater than 2, and the only even prime is 2 itself (Day 1, B9’s underlying fact) — so $p + 1$ cannot be prime.

Investigate further: This reuses B9’s “2 is the exceptional even prime” fact for the third distinct purpose in this pillar (Day 1’s B9, Day 4’s A4/B9, and now here) — a genuinely load-bearing fact worth having instantly available.

8. “Claim: for every positive integer n , $n^2 \geq 2n - 1$, with equality only at $n = 1$. ‘Proof’: $n^2 - 2n + 1 = (n - 1)^2 \geq 0$ always, with equality exactly when $n = 1$.” Is this proof valid?

Answer: Yes, fully valid.

Method: $n^2 - 2n + 1 = (n - 1)^2$ is an algebraic identity, and any real square is ≥ 0 , with equality iff the base is 0, i.e. $n = 1$. Rearranging $n^2 - 2n + 1 \geq 0$ gives exactly $n^2 \geq 2n - 1$, with equality iff $n = 1$.

Investigate further: Completing the square to turn an inequality into “something squared ≥ 0 ” is one of the cleanest and most reliable proof techniques available — worth recognising on sight as a strong candidate for being fully correct.

Section D Challenge

1. A student is asked to prove: “If n is even, then n^2 is even.” The student instead writes: “Suppose n^2 is even. Then $n^2 = 2m$ for some integer m . If n were odd, n^2 would be odd (Day 1, B5) — contradiction. So n is even.” Which best describes this attempt?

Answer: B) Incorrect — the student has proved the converse of the original statement instead.

Method: The original target is “ n even $\Rightarrow n^2$ even.” The student’s argument assumes n^2 even and concludes n even, which is “ n^2 even $\Rightarrow n$ even” — exactly the converse (Day 1), not the original. The student’s proof of that converse is itself valid (a correct proof by contradiction, previewing Day 7), but it proves the wrong target statement.

Investigate further: A perfectly valid proof can still be a wrong answer if it proves the converse instead of what was asked — this is the deepest trap in the whole pillar, combining Day 1 (converse), Day 3/7-preview (contradiction), and this sheet’s proof-critique skill all at once.

2. A student “proves” that $\sqrt{2}$ is rational as follows: “Suppose $\sqrt{2} = p/q$ in lowest terms. Then $2 = p^2/q^2$, so $2q^2 = p^2$, so p^2 is even, so p is even, so $p = 2r$. Then $2q^2 = 4r^2$, so $q^2 = 2r^2$, so q is even too. But p, q were in lowest terms, so they can’t both be even — contradiction. Hence $\sqrt{2}$ is rational.” Spot the flaw in the final line’s conclusion.

Answer: The contradiction actually proves $\sqrt{2}$ is *irrational*, not rational — every step up to the contradiction is a correct classical proof of irrationality.

Method: The argument assumes $\sqrt{2}$ is rational (expressible as p/q in lowest terms) and derives a genuine contradiction (both p, q forced even, contradicting “lowest terms”). In a proof by contradiction, reaching a contradiction from an assumption means that assumption is false — so the correct conclusion is that $\sqrt{2}$ is *not* rational, i.e. irrational. The student’s final sentence inexplicably states the opposite of what the contradiction actually establishes.

Investigate further: This is the single most famous proof by contradiction in elementary mathematics, and the flaw here isn’t in any line of algebra — it’s in correctly reading off what a completed contradiction actually proves, the exact skill Day 7’s toolkit will name explicitly.

3. “Reasoning: for every positive integer n , $n^2 + n$ is even; therefore, by strong induction, every polynomial in n with integer coefficients is even.” What is wrong with this reasoning, independent of whether the individual claim “ $n^2 + n$ is even” is itself true?

Answer: This is not a valid use of induction at all — induction proves a single indexed statement for all n , not an unrelated general claim about all polynomials from one example.

Method: Induction requires a base case and an inductive step for one specific statement $P(n)$, proving $P(n)$ holds for every n in its intended range. Verifying that one particular expression ($n^2 + n$) is even says nothing about a completely different, much broader claim (“every integer-coefficient polynomial in n is even”) — and that broader claim is in fact false, e.g. the polynomial n itself is odd when $n = 1$.

Investigate further: Invoking the *name* of a valid technique (“induction”) does not make an

argument valid — always check that the technique is actually being applied to the statement it's supposed to prove.

4. “Claim: a triangle has two equal angles if and only if it has two equal sides (isosceles); therefore, having two equal angles is logically equivalent to having a right angle.” What is wrong with this argument?

Answer: Non-sequitur — the conclusion about right angles doesn't follow from the premise about isosceles triangles at all.

Method: The premise (two equal angles $\iff$ isosceles) is a true geometric fact, but “therefore...having a right angle” introduces an entirely unrelated property with no logical connection established. A 70° - 70° - 40° isosceles triangle has two equal angles and no right angle, immediately breaking the claimed equivalence.

Investigate further: Unlike every other flawed argument on this sheet, the premise here isn't wrong at all — the entire flaw is a conclusion that simply doesn't connect to what came before it, the purest form of non-sequitur.

5. A student presents this argument: “Claim: if n is a positive integer with $n^2 - n + 41$ composite, then $n \geq 40$. Proof: we showed (Day 4, C1) that $n^2 + n + 41$ is prime for $n = 0, \dots, 39$ and composite at $n = 40$. Since $n^2 - n + 41$ is just $n^2 + n + 41$ with n replaced by $-n$, the same primality pattern applies, so $n^2 - n + 41$ is composite exactly when $n \geq 40$ too.” Identify precisely what is wrong with this argument, and give a specific counterexample.

Answer: Off-by-one substitution error: $n^2 - n + 41 = (n-1)^2 + (n-1) + 41$, so the “shifted” pattern actually starts breaking at $n = 41$, not $n = 40$. Counterexample to the student's claim: $n = 40$ gives $n^2 - n + 41 = 1601$, which is prime, not composite.

Method: Setting $m = n - 1$: $m^2 + m + 41 = (n-1)^2 + (n-1) + 41 = n^2 - 2n + 1 + n - 1 + 41 = n^2 - n + 41$, exactly matching the claim's expression. Since $m^2 + m + 41$ is prime for $m = 0, \dots, 39$ (Day 4, C1) and first composite at $m = 40$, the expression $n^2 - n + 41$ is prime for $n - 1 = 0, \dots, 39$ (i.e. $n = 1, \dots, 40$) and first composite at $n - 1 = 40$, i.e. $n = 41$ — not $n = 40$ as the student concluded. Directly: $n = 40$ gives $40^2 - 40 + 41 = 1600 - 40 + 41 = 1601$, which is prime (not divisible by any prime up to $\sqrt{1601} \approx 40$), directly contradicting the student's claim that composite values start at $n = 40$.

Investigate further: Substituting $-n$ for n inside a formula shifts *where* a pattern occurs, not just its shape — always re-derive the new threshold explicitly (here, via $m = n - 1$) rather than assuming the old one carries over unchanged. This closes the pillar's counterexample-and-critique arc by catching an error in reasoning *about* a fact from an earlier sheet, not just in a fresh claim.